

PC2193: Hall Effect in N- and P-type Germanium

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Abstract

This experiment investigated the Hall effect in n-type and p-type Germanium samples to characterize their fundamental electronic properties, specifically identifying dominant charge carrier types and determining carrier density (n) and carrier mobility (μ). By analyzing the Hall voltage (U_H) as a function of sample current and magnetic flux density, the primary charge carriers were confirmed as electrons for the n-type sample and holes for the p-type sample. The measured Hall coefficients (H) were $-(7.21 \pm 0.191) \times 10^{-3} \text{ m}^3/\text{C}$ for n-type and $+(7.09 \pm 0.188) \times 10^{-3} \text{ m}^3/\text{C}$ for p-type, corresponding to carrier densities of $(8.807 \pm 0.234) \times 10^{20} \text{ m}^{-3}$ and $(8.668 \pm 0.231) \times 10^{20} \text{ m}^{-3}$, respectively. Carrier mobility was evaluated using both Hall effect and magnetoresistance techniques, revealing a significant 58% disparity in p-type results ($0.282 \pm 0.007 \text{ m}^2/(\text{Vs})$ via Hall effect vs. $0.446 \pm 0.0097 \text{ m}^2/(\text{Vs})$ via magnetoresistance). This discrepancy is attributed to physical and statistical factors, including the Hall scattering factor (r) and the second-order nature of magnetoresistance path length increases. Furthermore, temperature-dependent measurements demonstrated a transition from extrinsic to intrinsic conduction, characterized by a "carrier inversion" effect in the p-type sample between 90°C and 100°C as high-mobility intrinsic electrons began to dominate the Hall signal.

1 Background & Objectives

The Hall effect, discovered in 1879 by Edwin Hall, describes the production of a transverse voltage (Hall voltage, U_H) across a conductor when placed in a perpendicular magnetic field. The hall effect has become an essential tool for solid state physicists for studying properties of semiconductors such as carrier density and carrier mobility. Knowing these characteristics of specific semiconductors is essential for their usage in all the electronic devices we find around us. The hall effect also has wider commercial uses in switches, joysticks, keyboards, digital storage devices and many more.

1.1 The Underlying Physics

As current flows in a conductor, the current carriers move with drift velocity v_d . When an external magnetic field B is applied, the moving charge carries feel a force F_L

$$\vec{F}_L = q(\vec{v}_d \times \vec{B}) \quad (1)$$

Due to this force the charge carriers shift to one edge of the conductor (in this case, the germanium sample). This in turn, creates a voltage difference between the two ends of the conductor which leads to an electric field E_H inside the conductor. This process continuous till the force due to E_0 perfectly balances F_L .

$$E_H = -(\vec{v}_d \times \vec{B}) \quad (2)$$

The voltage corresponding to this steady state is the hall voltage U_H . By expressing the Hall voltage as a function of I, rearranging and grouping some constants, we get:

$$H = \frac{U_H d}{BI} \quad (3)$$

Where d is the thickness of the sample and I is the current flowing through the sample. Using the hall coefficient we can obtain the carrier mobility μ and the carrier density n

$$\mu = H \cdot \sigma_0 \quad (4)$$

$$n = \frac{1}{eH} \quad (5)$$

Where σ_0 is the conductivity of the conductor at room temperature and e is the charge of an electron. The conductivity is calculated using the length L and cross sectional surface area A and the resistance at room temperature R_0 of the sample.

$$\sigma_0 = \frac{L}{R_0 A} \quad (6)$$

The carrier mobility can also be determined through an alternate method known as magnetoresistance. The resistance of the sample is a function of B according to the relationship:

$$R = R_0 + R_0(\mu B)^2 \quad (7)$$

Using Ohm's law ($V = IR$) we can convert this to

$$U_S = U_{S,0} + U_{S,0}(\mu B)^2 \quad (8)$$

Where $U_{S,0}$ is the sample voltage when magnetic field is 0. μ can also be determined from this relationship. The reason the hall effect is particularly useful for semiconductors is because, by doping, semiconductors can either have electrons or holes as the primary charge carriers. Due to opposite charges of the charge carriers, the hall voltage sign is different for the two types of semiconductors (p-type and n-type).

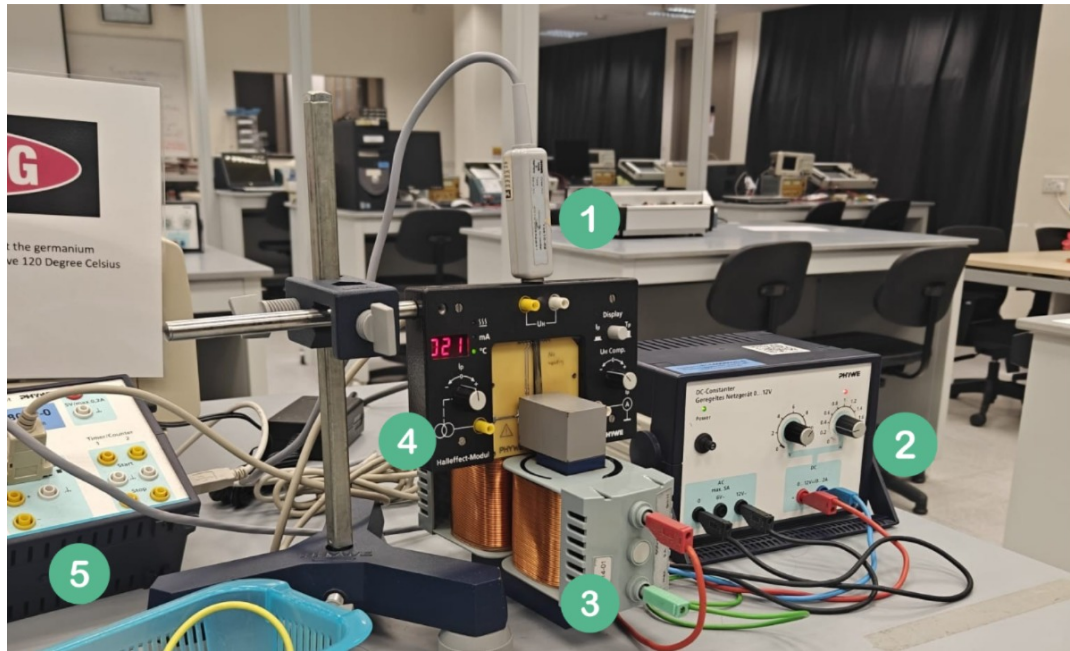
1.2 Objective

This experiment aims to observe the Hall effect in doped germanium samples. Key objectives include finding the Hall coefficient for both n and p-type samples and then using it to determine the carrier charge, the carrier mobility and the carrier density. The carrier mobility is also determined using method of magnetoresistance and then the value of μ is compared between the two methods. Lastly, the relationship between the hall voltage and sample temperature is studied.

2 Experimental Setup and Procedure

2.1 Apparatus

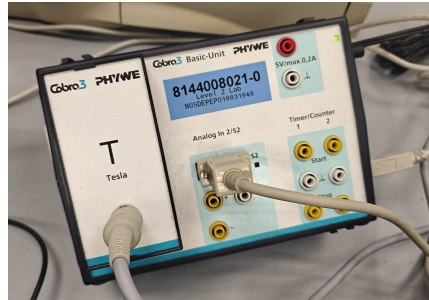
A labelled diagram of the setup is given below:



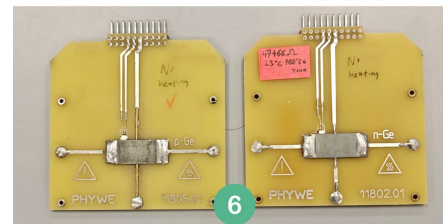
(a)



(b)



(c)



(d)

Figure 1: (a) All components involved in the experiment. (b) Control side of the Hall effect module. (c) The cobra 3 basic unit with Tesla module. (d) P and N-type germanium samples.

1. **Hall Probe:** The hall probe is placed in between the sample and the poles as seen and measures the B .
2. **Power Supply:** This provides power to the coils to generate the magnetic field. The magnetic field is adjusted by changing the voltage and current dials on the power supply unit. It also provides power to the hall effect module (4).
3. **600 turn coils, iron core and pole pieces:** They together provide the magnetic field perpendicular to the sample which is necessary for Hall effect.
4. **Hall effect module (HEM):** This houses the sample board and measures the hall voltage, allows current to flow through the sample and also has the heating coil to heat the sample. As seen in (b), the current flowing through the sample and U_H comp can be adjusted using this.
5. **Cobra3 Basic Unit and Tesla Module:** Acts as interface between all the components and the computer. Tesla module is connected to the Hall probe and measures B .
6. **Sample boards:** The contacts at the top slide into (4). The rectangular silver piece at the bottom is the germanium sample itself.

7. **Cobra3 Software:** Displays the data collected by all the components and helps take measurements.

2.2 Procedure and Experimental Methods

There were three separate experiments, each with some unique steps.

- The sample board and the hall probe were inserted into the HEM
- The HEM was then carefully lowered such that the sample was upright in between the pole pieces.

Experiment - 1 - Hall Voltage vs. Current

- Hall voltage(Y) and sample current(X) were chosen in the software as variables to be measured.
- Current was adjusted using the HEM from -30 mA to +30 mA in 5 mA steps.
- Current sweep was conducted at $B \approx 250$ mT and $B \approx 0$ mT and then the U_H values at 0 field were subtracted from the U_H values at 250 mT field to obtain the corrected U_H . U_H comp was not used.

Experiment - 2a - Sample Voltage vs. Magnetic Flux Density

- Sample voltage U_S (Y) and Magnetic flux density B (X) were chosen in the software as variables to be measured.
- Sample current was fixed to 30 mA. U_H comp unused.
- B was adjusted using the power supply from -300 mT to +300 mT in roughly 20 mT steps.
- At $B = 0$ polarity of the field was reversed by swapping the red and blue connections to the power supply.

Experiment - 2b - Hall Voltage vs. Magnetic Flux Density

- U_H (Y) and B (X) were chosen in the software as variables to be measured.
- Remaining procedure was the same as experiment 2a.

Experiment - 3 - Hall Voltage vs. Sample Temperature

- U_H (Y) and Sample temperature (X) were chosen in the software.
- I and B were fixed to 30 mA and 300 mT respectively.
- Heating was turned on using a button at the back of the HEM.
- Heating was shut before the sample reached 115°C.

There were more common steps for the experiments

- Each experiment was conducted 4 times (2 n-type and p-type runs) except experiment 3 which was conducted only twice (1 n-type and p-type run).
- Experiment 1 also had 2 $B = 0$ runs in addition to the 4 runs.
- Data was collected via key press for experiment 1 and 2. For experiment 3, readings were taken automatically every second.

3 Results and Analysis

3.1 Raw Data

The plots generated by the Cobra3 software (example shown below) were used to roughly confirm expected behaviour (for example for U_S vs. B curve for p-type should be a parabola).



Figure 2: Curve generated by Cobra3 software for U_S vs. B (blue line). The quadratic relationship between U_S and B is immediately visible. Other variables were also recorded and they are also shown in the graph.

In a number of cases, we took more than one reading at a particular value of I or B . This was done when the original reading was taken at a value very far away from the target value. For example, if we wanted to record U_H at 20 mA but recorded it at 16.5 mA, we retok the data point. The B value also sometimes jumped to double its value for a split second while measuring. These erroneous data points were removed before analysis. Averages of the two runs were considered for plotting the graphs.

3.2 Error Analysis

- Error for current measurement when current was held constant (in exp 2) was found to be negligible.
- Error in U_H was determined to be 0.5 mV. This value comes from fluctuations in U_H even after reaching steady state while taking measurements.
- Error in current for experiment 1 was determined to be 0.2 mA ($\approx 1.25\%$) as current is varied during this experiment.
- Error in d was approximated at 0.01mm(1%). Though, the dimensions of the sample were provided by the manufacturer and manually checked via ruler, it is completely possible for there to be manufacturing defects of a few micrometers which are undetectable using a ruler.
- Error of μ obtained via magnetoresistance has been obtained by considering the residuals from curve fitting.
- Error of the sample voltage is approximated at 0.001 V ($\approx 0.1\%$). This is because fluctuations in sample voltage were quite low during experimentation.

- Error of the magnetic flux density has been approximated at 3 mT ($\approx 2\%$). This comes from the fluctuation in B readings even after reaching steady state during the experiment.
- Errors of all other derived quantities have been obtained by considering propagation of errors in quadrature. The full calculations have been omitted, however, they are present on my laptop.
- For slope errors, relative error with respect to the mean of absolute values has been considered rather than every single data point.

3.3 Experiment - 1

The linear relationship between U_H and I is reinforced by fitting curves using linear regression.

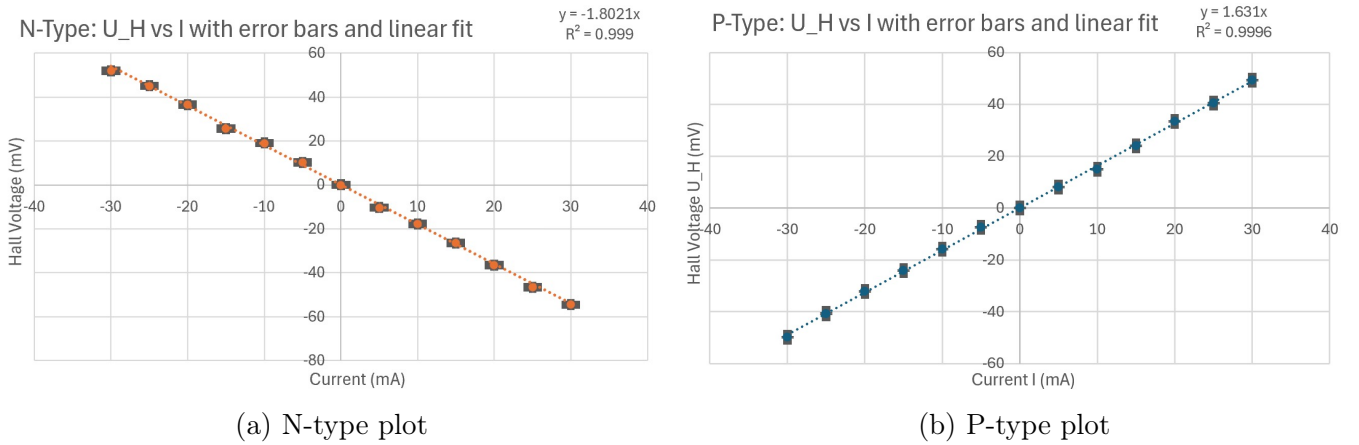


Figure 3: U_H dependence on I for n-type and p-type germanium samples

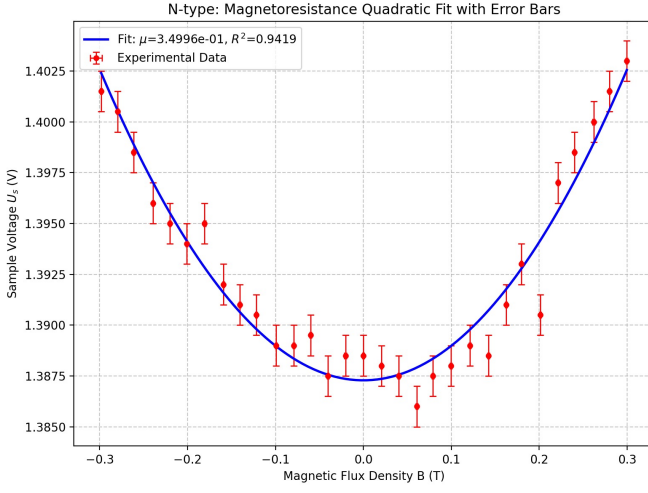
From the plots:

1. N-type slope = -1.8021 ± 0.0411 V/A
2. P-type slope = 1.631 ± 0.037 V/A

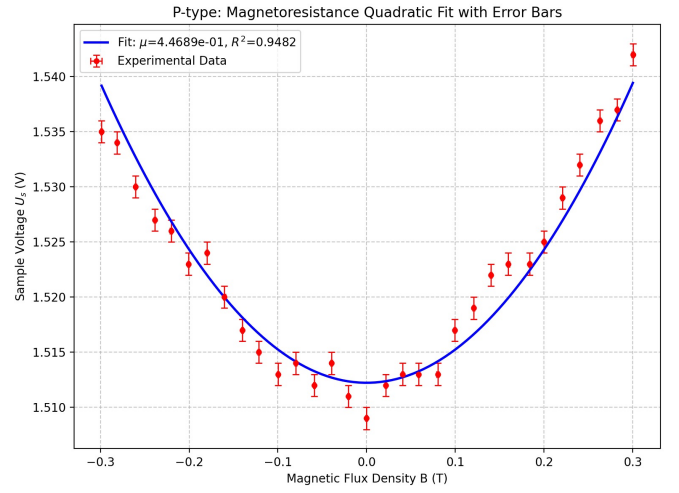
The signs of the slopes confirm the charge of the carriers in the samples. The negative slope for N-type confirms electrons are the primary charge carriers whereas the positive slope for P-type confirms that holes are the primary charge carriers. Using these slopes and equation (3), we can calculate the hall coefficient with $d = 0.001\text{m}$ and $B = 250$ mT. However, we will use data from experiment 2b to calculate the hall coefficient instead.

3.4 Experiment - 2a

The U_S vs B data was fit according to equation (8). The baseline voltage $U_{S,0}$ was taken as the sample voltage corresponding to $B \approx 0$ mT.



(a) N-type plot



(b) P-type plot

Figure 4: U_S dependence on B for n-type and p-type germanium samples

The high R^2 values for both plots reinforce the quadratic relationship between the resistance (and hence sample voltage) and B as shown in equation (7). The fitting yielded the values of the carrier mobility μ .

1. $\mu_{mn} = 0.349 \pm 0.0081 \text{ m}^2/(\text{Vs})$
2. $\mu_{np} = 0.446 \pm 0.0097 \text{ m}^2/(\text{Vs})$

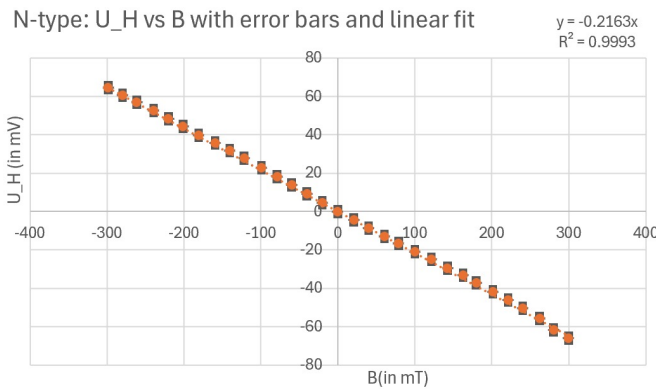
We also obtain the values of R_0 by considering $U_{S,0}$.

$$R_0 = \frac{U_{S,0}}{I} \quad (9)$$

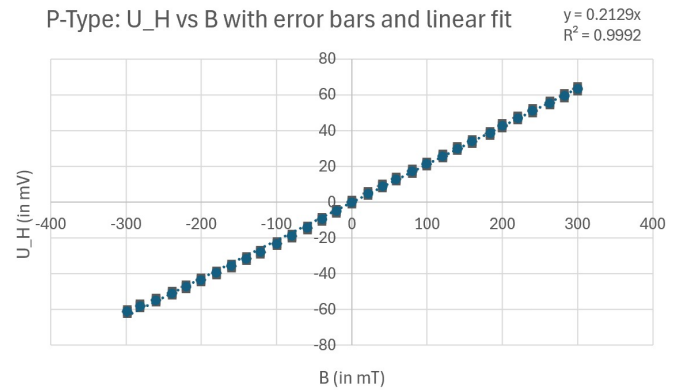
Where $I = 0.030 \text{ mA}$

1. **N-type** $R_0 = 46.27 \pm 0.033 \Omega$
2. **P-type** $R_0 = 50.3 \pm 0.033 \Omega$

3.5 Experiment - 2b



(a) N-type plot



(b) P-type plot

Figure 5: U_H dependence on B for n-type and p-type germanium samples

The extremely high R^2 values of both plots reinforce the linear relationship between U_H and B as shown in equation (3). From the plots,

1. **N-type slope:** $-0.2163 \pm 0.0053 \text{ V/T}$

2. **P-type slope:** $0.2129 \pm 0.0052 \text{ V/T}$

The difference in sign in the slopes yet again confirms negatively charged electrons are the primary charge carriers in the N-type sample whereas the positively charged holes are the charge carriers in the P-type sample. The slopes represent the value of the ratio $\frac{U_H}{B}$ which we can directly plug into equation (3) to find the hall coefficients.

1. **N-type, $H_n = (-7.21 \pm 0.191) \times 10^{-3} \text{ m}^3/\text{C}$**

2. **P-type, $H_p = (7.09 \pm 0.188) \times 10^{-3} \text{ m}^3/\text{C}$**

Using the hall coefficient we can determine the carrier mobility and the carrier density using equations (4) and (5). Values of σ_0 for both samples are obtained by using R_0 (obtained in previous experiment) and the dimensions of the sample using equation (6). Carrier mobility values:

1. **N-type, $\mu_{hn} = 0.312 \pm 0.008 \text{ m}^2/(\text{Vs})$**

2. **P-type, $\mu_{hp} = 0.282 \pm 0.007 \text{ m}^2/(\text{Vs})$**

As apparent, these values differ significantly from the μ values found using magnetoresistance (refer to Discussion) Carrier density values:

1. **N-type, $n_n = (8.807 \pm 0.234) \times 10^{20} \text{ m}^{-3}$**

2. **P-type, $n_p = (8.668 \pm 0.231) \times 10^{20} \text{ m}^{-3}$**

3.6 Experiment - 3

A simple scatter plot of the U_H and T_p data was plotted for both p-type and n-type.

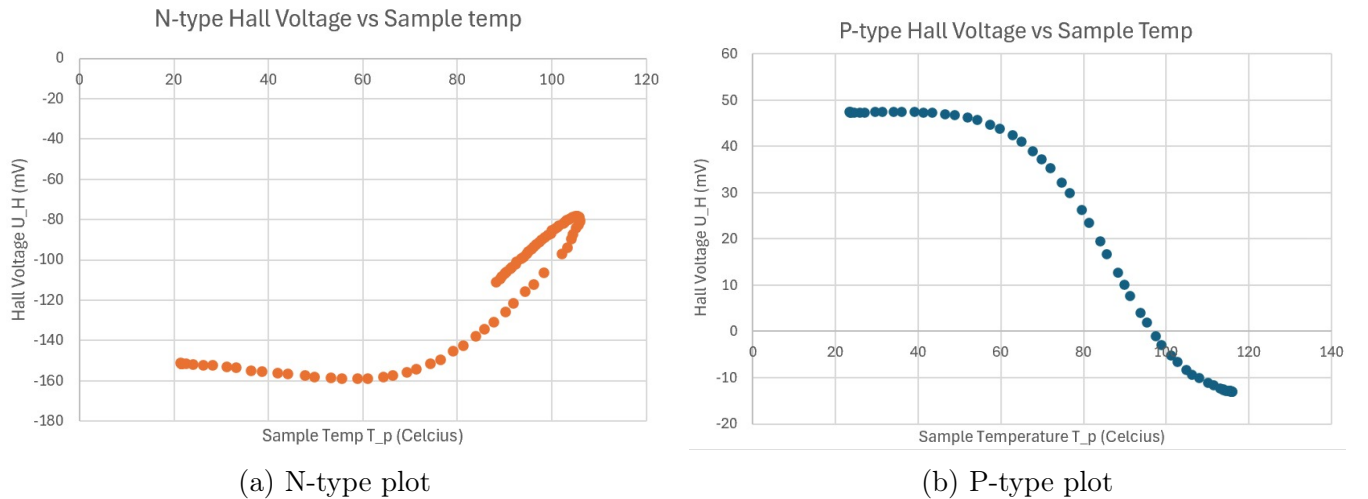


Figure 6: U_H dependence on T_p for n-type and p-type germanium samples

The tail in the plot for the n-type sample is because the measurement was running even after heating was stopped and thus the tail represents a part of the cooling curve of U_H .

The temperature dependence of the Hall voltage (U_H) for both germanium samples provides a clear visualization of the transition from extrinsic to intrinsic conduction as thermal energy increases.

For the n-type sample, the magnitude of the negative Hall voltage decreased with rise in temperature. This is due to the thermal generation of intrinsic electron-hole pairs, which increases

the total carrier concentration (n); since U_H is inversely proportional to n , the additional carriers effectively suppress the Hall potential.

In contrast, the p-type sample exhibited a "carrier inversion" effect. At lower temperatures, the Hall voltage remained positive, confirming holes as the majority carriers. However, as the temperature approached approximately $90^\circ\text{C} - 100^\circ\text{C}$, U_H was observed to decrease rapidly, pass through zero, and become negative. This occurs because the thermally generated intrinsic electrons have a significantly higher mobility than holes in Germanium. Consequently, these high-mobility electrons begin to dominate the Hall signal even before the sample becomes fully intrinsic, eventually dictating a negative Hall voltage at high temperatures.

4 Discussion

4.1 Summary of Experimental Results

The following table summarizes the measured and calculated physical parameters for both the n-type and p-type germanium samples.

Table 1: Summary of Experimental Results

Variable	n-type Germanium	p-type Germanium
Charge Carrier Type	Negative (Electrons)	Positive (Holes)
Hall Coefficient H (m^3/C)	$-(7.21 \pm 0.191) \times 10^{-3}$	$+(7.09 \pm 0.188) \times 10^{-3}$
Carrier Density n, p (m^{-3})	$(8.807 \pm 0.234) \times 10^{20}$	$(8.668 \pm 0.231) \times 10^{20}$
Magnetoresistance Mobility μ_m (m^2/Vs)	0.349 ± 0.0081	0.446 ± 0.0097
Hall Mobility μ_h (m^2/Vs)	0.312 ± 0.008	0.282 ± 0.007
Mobility ratio (μ_m/μ_h)	1.118	1.582

4.2 Interpretation of Results

4.2.1 Charge Carrier Type and Sign

The polarity of the Hall voltage relative to the direction of the sample current I and magnetic field B allowed for the identification of the majority charge carriers. For the p-type sample, a positive Hall voltage was observed, confirming that holes are the primary carriers. Conversely, the n-type sample yielded a negative Hall voltage, identifying electrons as the majority carriers. These results align with the theoretical behavior of doped semiconductors under the Lorentz force.

4.2.2 Hall Coefficient and Carrier Density

The calculated Hall coefficients (R_H) are on the order of $10^{-3} \text{ m}^3/\text{C}$, which is typical for moderately doped semiconductors like Germanium. The resulting carrier densities ($n \approx 10^{20} \text{ m}^{-3}$) confirm that the samples are extrinsically doped since these densities are much higher than the carrier density of undoped germanium ($\approx 10^{19}$). The slight discrepancy between the n-type and p-type densities suggests variations in the doping concentrations provided by the manufacturer.

4.2.3 Magnetoresistance and Mobility

- **Mobility Values:** The p-type sample exhibited a higher mobility ($\approx 0.447 \text{ m}^2/\text{Vs}$) compared to the n-type sample ($\approx 0.350 \text{ m}^2/\text{Vs}$). This is expected since the electrons have higher

mobility than holes.

- **Fit Accuracy:** High R^2 values (> 0.94) for both samples validate the quadratic model for low-field magnetoresistance in semiconductors.
- **Physical Meaning:** Mobility represents how quickly a carrier can move through the crystal lattice when an electric field is applied. The measured values are within the expected range for Germanium at room temperature..

4.3 Leading sources of error

The two leading sources of uncertainty in this experiment include:

1. The drift in B values contribute significantly to the uncertainty to the Hall coefficients which in turn propagates to all the other variables.
2. The uncertainty in the dimensions of the sample contribute directly to the increase in uncertainty in the values of carrier mobility (calculation using hall effect), the resistance and the conductivity.

4.4 Disparity in Mobility Values

A notable observation in the results is the numerical difference between the mobility calculated from the Hall Coefficient (μ_h) and that derived from the Magnetoresistance fit (μ_m). For n-type the disparity is around 11% but for p-type the disparity is very significant at around 58%. This disparity arises from several physical and statistical factors:

1. **The Hall Scattering Factor (r):** The relationship between the measured Hall mobility (μ_H) and the actual drift mobility (μ_d) is defined by $\mu_H = r\mu_d$. The factor r depends on the dominant scattering mechanism within the crystal lattice. For Germanium at room temperature, where acoustic phonon (lattice) scattering dominates, theory predicts $r \approx 1.18$. Consequently, μ_h is typically expected to be higher than μ_d .
2. **Statistical Weighting of Velocity:**
 - **Hall Effect:** Measures the average deflection of carriers under the Lorentz force, which is essentially a first-order effect proportional to the mean relaxation time $\langle\tau\rangle$.
 - **Magnetoresistance:** This is a second-order effect ($\propto B^2$) that depends on the *variance* in carrier relaxation times across the energy distribution. Faster-than-average and slower-than-average carriers contribute differently to the magnetoresistive increase in path length, often leading to a μ_p value that differs from μ_h .

5 Conclusion

In this experiment, the Hall Effect was successfully used to characterize two Germanium semiconductor samples. We identified the p-type sample as having positive charge carriers and the n-type sample as having negative charge carriers. The Hall coefficients were determined to be $-(7.21 \pm 0.191) \times 10^{-3} \text{ m}^2/(\text{Vs})$ and $+(7.09 \pm 0.188) \times 10^{-3} \text{ m}^2/(\text{Vs})$ for the n- and p-type samples respectively. The hall coefficient was used to then calculate carrier mobility and carrier density. Carrier mobility was also calculated using magnetoresistance and the discrepancy in values using the two methods was explored. The high correlation in our fits and the alignment with theoretical expectations demonstrate the validity of the Hall Effect as a robust tool for semiconductor analysis. Future iterations can implement better ways to measure the dimensions of the sample and more

stable readings of B . More stable readings of B can be achieved by fixing the HEM and hall probe in the form of a locked sliding mechanism such that the hall probe orientation does not change easily during measurement due to external vibrations etc.

6 References

1. PC2193 Experimental Physics Lab Manual: Hall Effect.
2. PHYWE SYSTEME GMBH, "Hall effect in n- and p-germanium" P2530116.
3. A. Kittel, C. (2004). Introduction to Solid State Physics. Wiley.

7 AI Usage Declaration

I declare that I have used AI in the preparation of this assignment. Prompt: "create a full latex template code complete with hyperlinks in the contents page to all the different sections with structure given in the images. Add a way to check the word count." (the images of DOs and DONTs and report structure were uploaded along with the prompt) Response: Gemini created a full LaTeX code template which I used to make the report. In general, AI was used for help with LaTeX codes, templates and troubleshooting when LaTeX code did not compile. AI was also used to troubleshoot when python code for analysing experiment 2a data failed to compile.